

Soon Jie, Elijah

Singapore
elijahjie@gmail.com | +65 90044635



PROFESSIONAL SUMMARY

Cybersecurity and Systems Engineering student at the Singapore University of Technology and Design (SUTD) with strong expertise in Windows internals, Sysmon telemetry analysis, and kernel-level event tracing. Experienced in full-stack development and machine learning system design. Proven ability to conduct deep technical research, validate system behaviour through debugging and experimentation, and present structured findings to defence stakeholders.

TECHNICAL SKILLS

Programming: Python, Java, JavaScript, React, Node.js

Cybersecurity & Systems:

- Windows Sysmon telemetry analysis
- Black-box security testing
- Capture The Flag (CTF) participation (binary exploitation, log analysis, system investigation)

Data & Machine Learning:

- Machine learning model development
 - Data preprocessing and feature engineering
 - Full-stack ML web application integration
-

WORK EXPERIENCE

Software Developer

Ministry of Defence, Singapore
Apr 2024 – Jul 2024

- Designed and implemented automated contract monitoring systems with secure login validation and permission controls
- Developed automated email notification pipelines for contract lifecycle tracking
- Enhanced data integrity mechanisms and structured access control frameworks

Cybersecurity Research Attachment

Ministry of Defence, Singapore

Sept 2025 – Jan 2026

- Conducted deep technical analysis of Sysmon internals, tracing event generation from kernel callbacks to EVTX storage
- Investigated structural weaknesses and potential bypass techniques within Sysmon telemetry pipelines
- Reproduced documented evasion techniques in controlled environments to validate detection gaps
- Presented structured technical findings and risk implications to senior stakeholders
- Conducted research on RF signal behaviour, Wi-Fi protocols, and drone communication systems to understand system-level vulnerabilities

Freelance Developer

Etiqua Insurance

Jan 2025

- Developed automated budgeting and financial tracking tools
- Built a secure algorithm-based lucky draw system ensuring fairness, transparency, and auditability

SELECTED PROJECTS

Windows Sysmon Deep Telemetry Research

- Analysed Sysmon data collection mechanisms, including kernel callbacks, ETW providers, and minifilter drivers
- Reverse-tested configuration filtering logic and evaluated detection reliability
- Mapped the full event pipeline from kernel trigger to event log storage
- Validated telemetry sources using fltmc, registry analysis, PerfMon, and WinDbg
- Structured research into a formal technical presentation for defence stakeholders

Machine Learning-Based Food Wastage Prediction System

- Developed a machine learning model integrated into a full-stack web application
 - Built data pipeline, preprocessing workflow, and deployment architecture
 - Recognised as Best Project among 200 competing teams
-

Full-Stack Hotel Booking Platform

- Built React frontend and Node/Express backend
 - Integrated MongoDB/MySQL database architecture
 - Implemented dynamic search, filtering, and booking logic
 - Designed scalable system architecture for loyalty integration
-

EDUCATION

Singapore University of Technology and Design (SUTD)

Bachelor of Science (Honours) in Computer Science and Design, Minor in Artificial Intelligence
2023 – Present

CERTIFICATIONS

Cisco Certified Network Associate (CCNA)

AWARDS

VEX Robotics World Champion – 2017
Represented Singapore internationally